

ATLAS SYSTEM HANDOVER DOSSIER

Mission-Ready State Report | July 3, 2026

1. Executive Summary

This dossier contains the complete context of Project Atlas as of July 3, 2026. The system is a functional prototype for a Master Flightline Clock, designed for competition aviation environments. It has moved from basic JSON configuration to a professional, database-driven appliance architecture running on a Mac Mini M4.

2. Current Project State

Last Stable Version: v1.5.0

Key Features Implemented:

- SQLite Persistence for Heats & Settings.
- 6-Row Adaptive LED Rendering (128x64 and 64x64 profiles).
- Experimental "IntensityControl" for LED dimming.
- Custom High-Contrast Cockpit GUI with Navigation Tab system.
- Graceful System Shutdown/Restart logic.

3. File System Snapshot

```
Project_Atlas/  
├── src/  
│   ├── main.py (Entry point/Wiring)  
│   ├── core/  
│   │   ├── models.py (Data structures/Display model)  
│   │   ├── database.py (SQLite CRUD logic)  
│   │   └── engine.py (10Hz State Machine)  
│   └── gui/  
│       ├── windows.py (Operator Console UI)  
│       └── themes.py (Custom Styles/Colors)
```

```
| |— led_simulator.py (Public Display Sim)
| |— fonts.py (Custom Bitmap font engine)
|— docs/ (SDS, Charter, URS)
|— run_atlas.sh (Main Launch Script)
|— atlas.db (Primary Data Store)
```

4. Component Breakdown

Module	Responsibility
AtlasEngine	Calculates UTC-based countdown phases (P0-P4).
AtlasDatabase	Handles schema versioning and data persistence.
OperatorWindow	Multi-tab control panel for schedule and settings.
AtlasLEDSimulator	Dynamic P-series matrix simulation with brightness scaling.

5. Pending Tasks (Post-Handover)

- **Hardware Handshake:** Implement network communication to physical LED controllers.
- **Documentation Finish:** Finalize the Interface Control Document (ICD).
- **Traceability:** Finalize the Traceability Matrix (TM-001).
- **Installer:** Create a .dmg or automated setup script for dedicated Mac Mini deployment.

6. User Preferences & Tone

The system is maintained for **Peter Lochhead Hepburn**. Tonal preference: Casual, humorous, technical, and concise. Technical background: IT Director, GAMP 5, hardware enthusiast (Jeep TJs, Raptor). Location: Adelaide (Strathalbyn), South Australia.